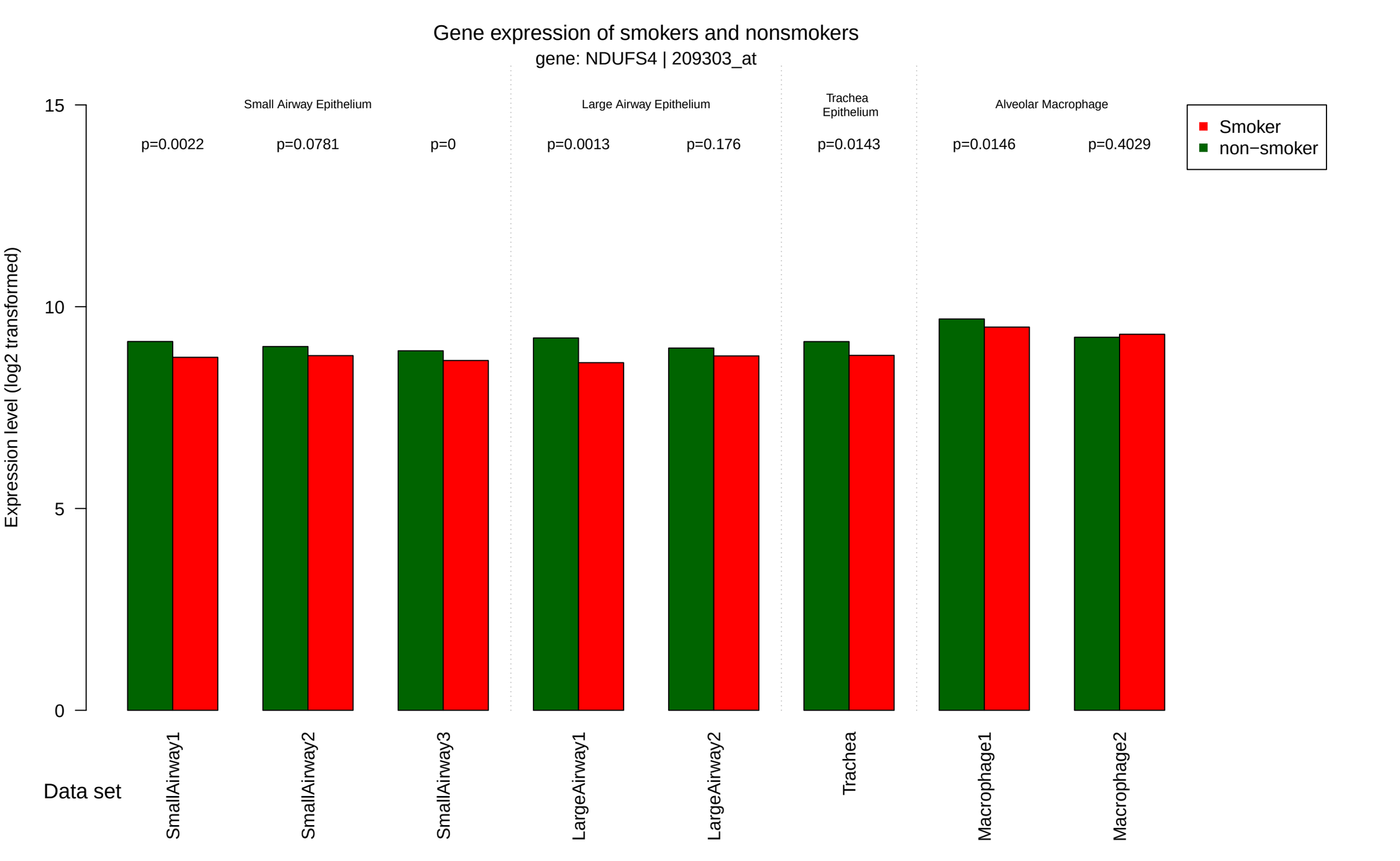
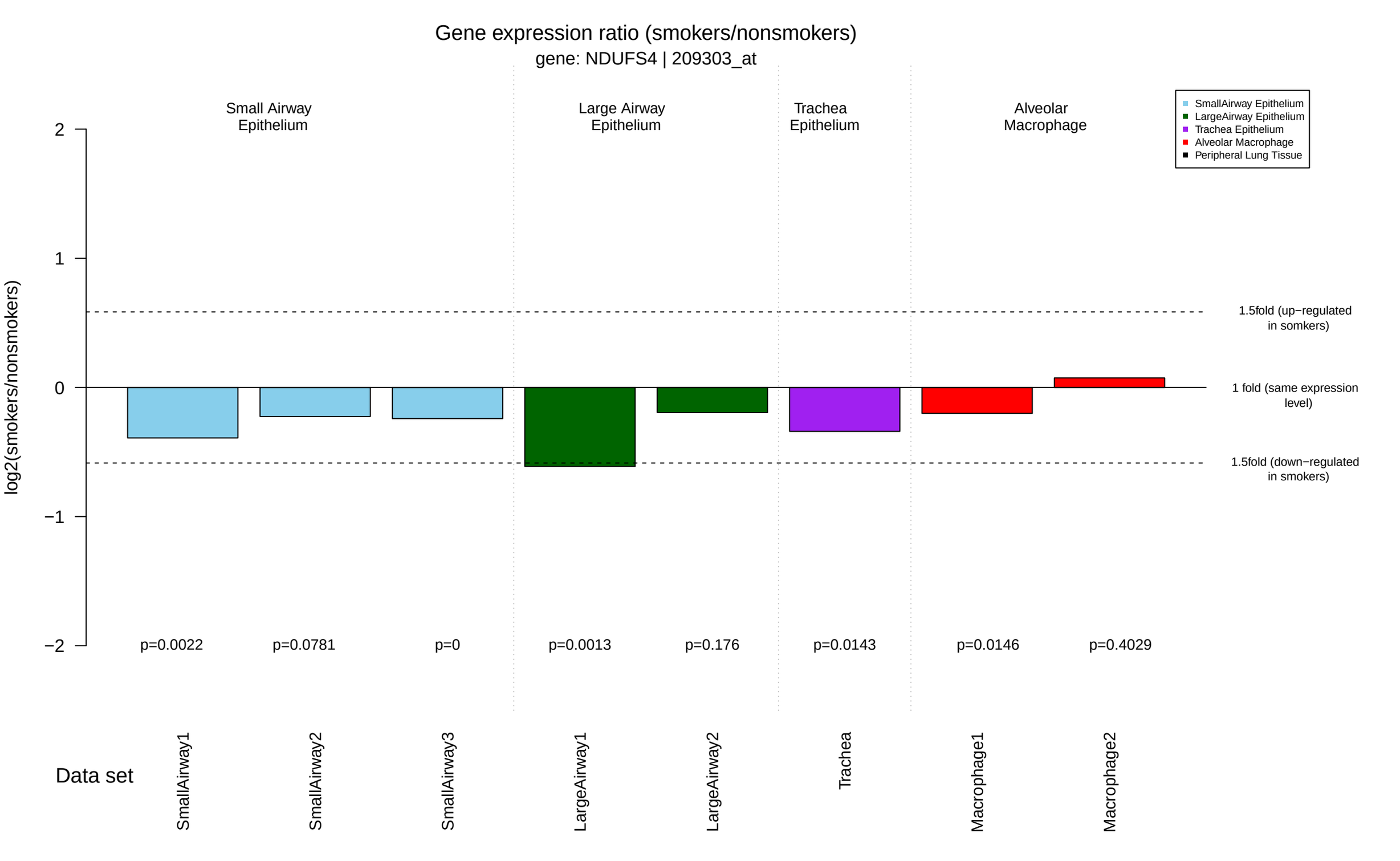
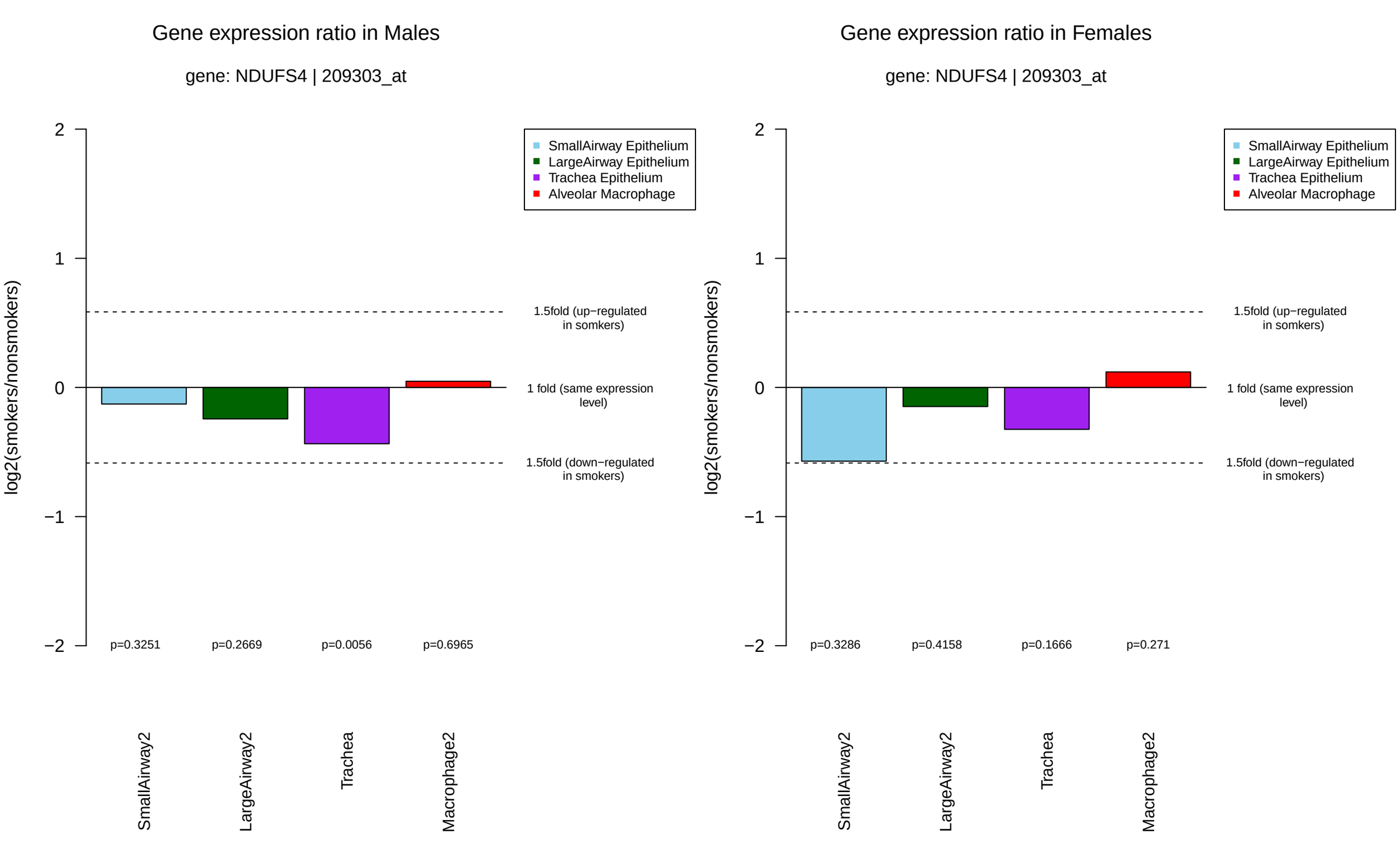
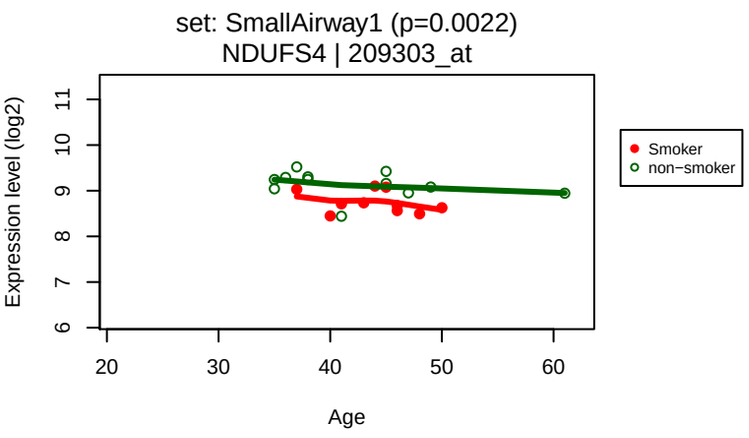
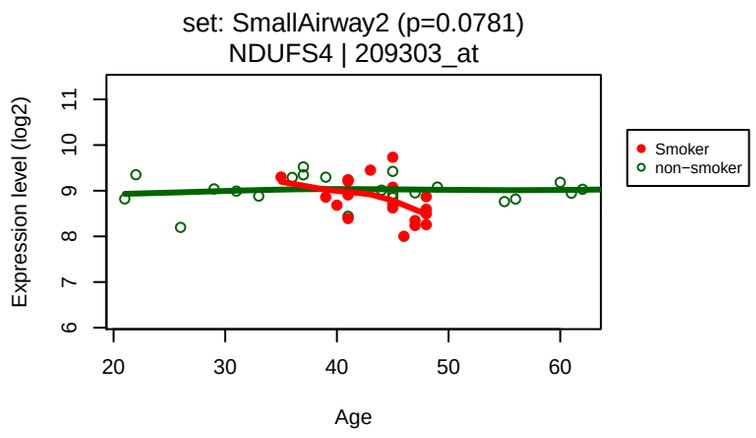
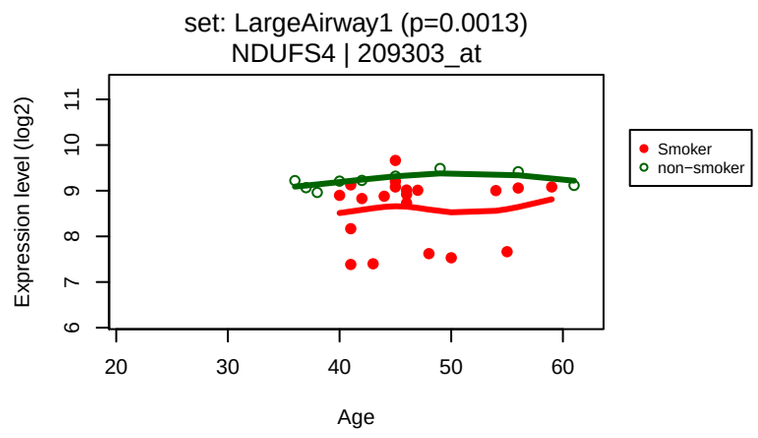

Gene expression of smokers and nonsmokers
gene: NDUFS4 | 209303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0022
p=0.0781
p=0
p=0.0013
p=0.176
p=0.0143
p=0.0146
p=0.4029
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

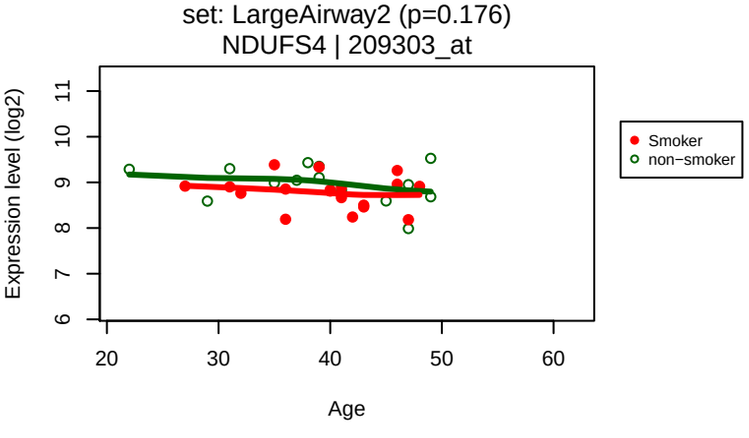

Gene expression ratio (smokers/nonsmokers)
gene: NDUFS4 | 209303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0022
p=0.0781
p=0
p=0.0013
p=0.176
p=0.0143
p=0.0146
p=0.4029
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

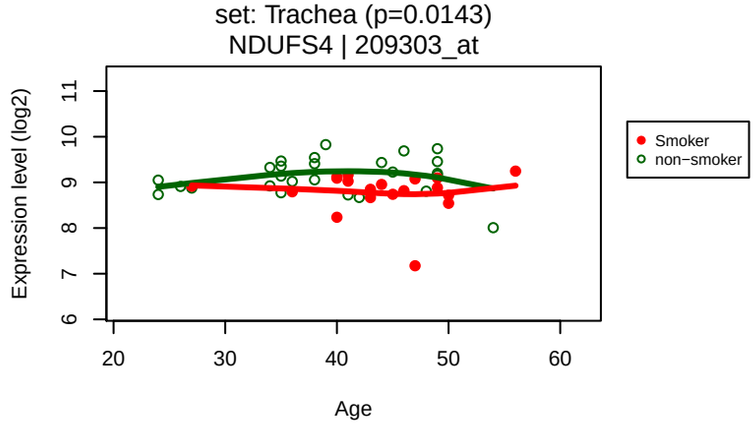

Gene expression ratio in Males
gene: NDUFS4 | 209303_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3251
p=0.2669
p=0.0056
p=0.6965
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NDUFS4 | 209303_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3286
p=0.4158
p=0.1666
p=0.271
SmallAirway2
LargeAirway2
Trachea
Macrophage2

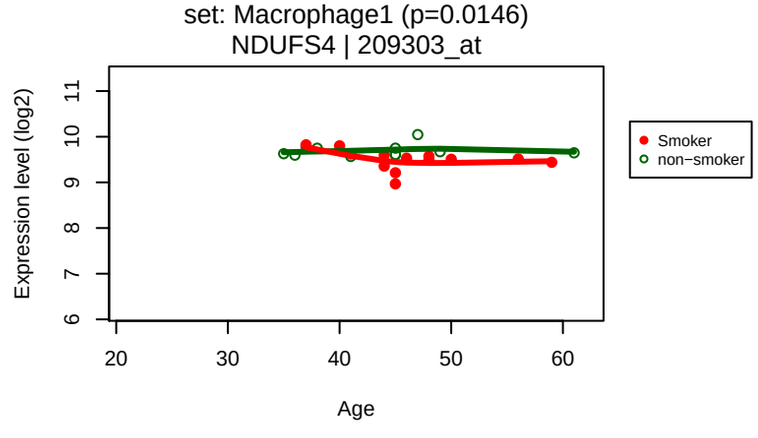

set: SmallAirway1 (p=0.0022)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker

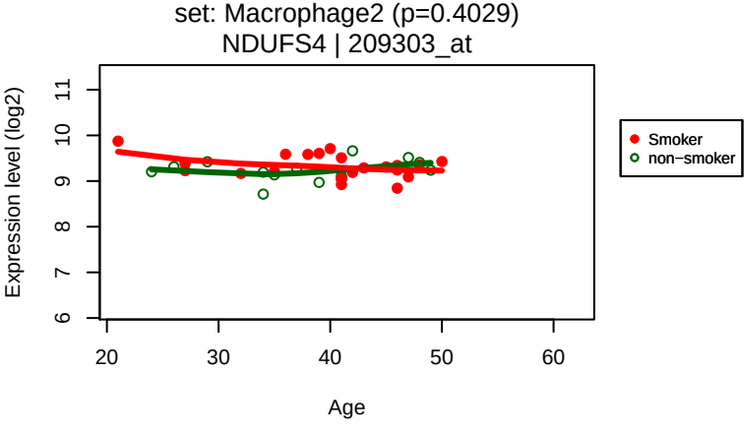

set: SmallAirway2 (p=0.0781)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker

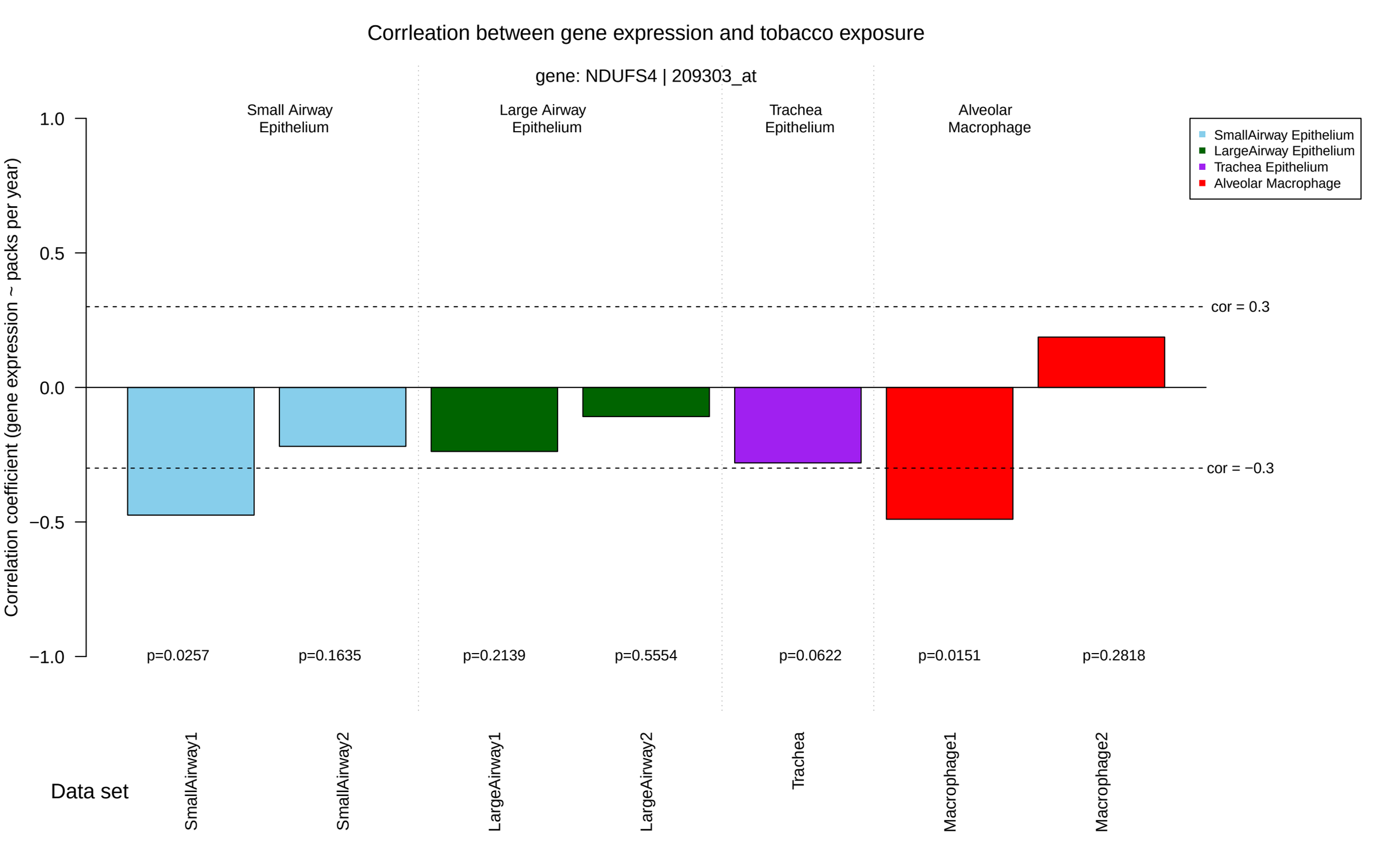

set: LargeAirway1 (p=0.0013)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.176)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0143)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0146)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4029)
NDUFS4 | 209303_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NDUFS4 | 209303_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0257
p=0.1635
p=0.2139
p=0.5554
p=0.0622
p=0.0151
p=0.2818
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2